

Strong Distance of Complete Bipartite Graphs

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Abstract

For two vertices u, v in a strong digraph D , the strong distance $sd(u, v)$ between u and v is the minimum size of a strong subdigraph of D containing u and v . For a vertex v of D , the strong eccentricity $se(v)$ is the maximum strong distance between v and all vertices in D . The strong radius $srad(D)$ is the minimum strong eccentricity among all the vertices of D , and the strong diameter $sdiam(D)$ is the maximum strong eccentricity among all the vertices of D . The lower orientable strong radius $srad(G)$ of a graph G is the minimum strong radius over all strong orientations of G . The upper orientable strong radius $SRAD(G)$ of a graph G is the maximum strong radius over all strong orientations of G . The lower orientable strong diameter $sdiam(G)$ of a graph G is the minimum strong diameter over all strong orientations of G . The upper orientable strong diameter $SDIAM(G)$ of a graph G is the maximum strong diameter over all strong orientations of G . This paper provides the lower and upper orientable strong radius and diameter of complete bipartite graphs.

Keywords: Strong distance, complete bipartite graph, strong radius, strong diameter.

1 Introduction

In [2], they defined *strong distance* $sd(u, v)$ between u and v for two vertices u, v in a strong digraph D as is the minimum size of a strong subdigraph of D containing u and v . Follow the definitions in [2], the *strong eccentricity* $se(v)$ of a vertex v in D is

$$se(v) = \max\{sd(v, x) : x \in V(D)\};$$

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and the *strong radius* $srad(D)$ of D is

$$srad(D) = \min\{se(v) : v \in V(D)\};$$

while the *strong diameter* $sdiam(D)$ of D is

$$sdiam(D) = \max\{se(v) : v \in V(D)\}.$$

For a connected graph G , we define the *lower orientable strong radius* $srad(G)$ of G by $srad(G) = \min\{srad(D) : D \text{ is a strong orientations of } G\}$; while the *upper orientable strong radius* $SRAD(G)$ of G by $SRAD(G) = \max\{srad(D) : D \text{ is a strong orientations of } G\}$. We also define the *lower orientable strong diameter* $sdiam(G)$ of G by $sdiam(G) = \min\{sdiam(D) : D \text{ is a strong orientations of } G\}$; while the *upper orientable strong diameter* $SDIAM(G)$ of a graph G by $SDIAM(G) = \max\{sdiam(D) : D \text{ is a strong orientations of } G\}$.

In [1] they discussed the strong radius and strong diameter of complete graphs, and [2] gave upper bound for the strong diameter of a strong oriented graph. Here we turn our attention to the complete bipartite graphs.

2 Orientable strong radius

The girth $g(G)$ of a graph G with cycles is the length of a smallest cycle in G . Lemma 1 is used in the proof of the lower orientable strong radius of the complete bipartite graphs.

Lemma 1 Let $G = (V, E)$ be a connected graph with n vertices and D be an orientation of G . Then $srad(D) \geq g(G)$.

Proof: Let $V(G) = \{v_i : \text{for } 1 \leq i \leq n\}$, and $C = \{v_1, v_2, \dots, v_l, v_1\}$ be a cycle whose length is $l = g(G)$. Oriente the edges of G cyclically so that (v_j, v_{j+1}) is an arc in D for $1 \leq j \leq l-1$ and (v_l, v_1) is an arc in D . Then there exists a strong distance $sd(v_r, v_s) = l$ for $1 \leq r, s \leq l$, and $r \neq s$. This implies $srad(D) \geq l = g(G)$. $\square$

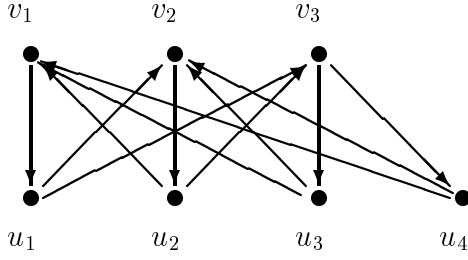


Figure 1: An orientation D of $K_{3,4}$ such that $\text{srad}(D) = 4$.

Theorem 1 Let $2 \leq m \leq n$. Then $\text{srad}(K_{m,n}) = 4$.

Proof: Let $V = \{v_1, v_2, \dots, v_m\}$, $U = \{u_1, u_2, \dots, u_n\}$ be two partite sets of $K_{m,n}$. For any strong orientation of $K_{m,n}$, since $g(K_{m,n}) = 4$, by Lemma 1, we have $\text{srad}(G) \geq 4$. Consider an orientation D_1 of $K_{m,n}$ such that (v_i, u_i) is an arc in D_1 for $1 \leq i \leq m$; (v_m, u_j) is an arc in D_1 for $m+1 \leq j \leq n$, and oriente all other edges from U to V . Then $\text{srad}(D_1) = 4$, which implies $\text{srad}(K_{m,n}) = 4$. $\square$

Figure 1 shows an orientation D of $K_{3,4}$ such that $\text{srad}(D) = 4$.

Next theorem give the lower bound of the upper orientable strong radius of the complete bipartite graphs.

Theorem 2 Let $2 \leq m \leq n$. Then

$$\text{SRAD}(K_{m,n}) \geq \begin{cases} m+1 & \text{if } m \text{ is odd,} \\ m+2 & \text{if } m \text{ is even.} \end{cases}$$

Proof: Let $V = \{v_1, v_2, \dots, v_m\}$, $U = \{u_1, u_2, \dots, u_n\}$ be two partite sets of $K_{m,n}$.

Case 1: Let $m = n$. Consider an orientation D_1 of $K_{m,n}$ such that (v_i, u_{i+1}) is an arc in D_1 for $1 \leq i \leq m-1$, (v_k, u_l) is an arc in D_1 for $2 \leq k \leq m$, $l < k$, and oriente all other edges from U to V . Then

$$\text{srad}(D_1) = \begin{cases} m+1 & \text{if } m \text{ is odd,} \\ m+2 & \text{if } m \text{ is even,} \end{cases}$$

which implies

$$\text{SRAD}(K_{m,n}) \geq \begin{cases} m+1 & \text{if } m \text{ is odd,} \\ m+2 & \text{if } m \text{ is even.} \end{cases}$$

Examples of the orientation for $K_{5,5}$ and $K_{4,4}$ are shown in Figure 2 and 3.

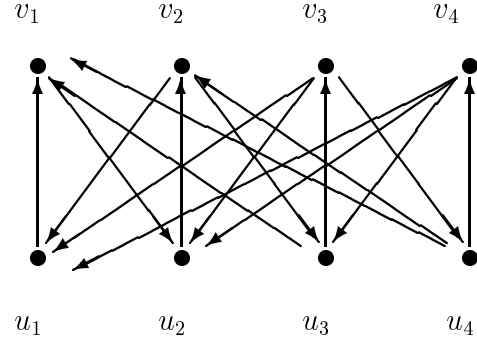


Figure 3: An orientation to reach lower orientable strong radius of and $K_{4,4}$.

Case 2: Let $m \neq n$. Consider an orientation D_2 of $K_{m,n}$ such that (v_i, u_{i+1}) is an arc in D_2 for $1 \leq i \leq m$, (v_m, u_j) is an arc in D_2 for $m+2 \leq j \leq n$, (v_k, u_l) is an arc in D_2 for $2 \leq k \leq m$, $l < k$, and oriente all other edges from U to V . Then

$$\text{srad}(D_2) = \begin{cases} m+1 & \text{if } m \text{ is odd,} \\ m+2 & \text{if } m \text{ is even,} \end{cases}$$

which implies

$$\text{SRAD}(K_{m,n}) \geq \begin{cases} m+1 & \text{if } m \text{ is odd,} \\ m+2 & \text{if } m \text{ is even.} \end{cases}$$

Examples of the orientation for $K_{5,6}$ and $K_{4,5}$ are shown in Figure 4 and 5. $\square$

3 Orientable strong diameter

In this section, we provide the upper orientable strong diameter and lower orientable strong diameter for the complete bipartite graphs.

Lemma 2 For any orientation D of $K_{m,n}$, $\text{sd}(u, v)$ is even for $u, v \in V(D)$.

Proof: Let $V = \{v_1, v_2, \dots, v_m\}$, $U = \{u_1, u_2, \dots, u_n\}$ be two partite sets of $K_{m,n}$. Let r, s be two vertices in D , which are in the same partite set. Let P_1 be a shortest directed path from r to s , P_2 be a shortest directed path from s to r . We know that both $|P_1|$ and $|P_2|$ are even.

Case 1: P_1 and P_2 have even or no common edge. Then $\text{sd}(r, s) = |P_1| + |P_2|$ which is even.

Case 2: P_1 and P_2 have odd common edge. Without loss of generality, assume that there is only one common edge (say xy).

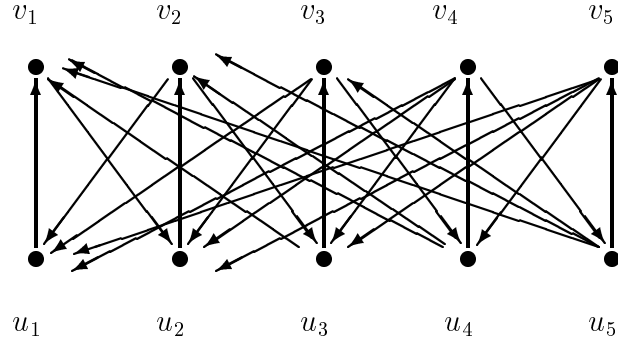


Figure 2: An orientation to reach lower orientable strong radius of $K_{5,5}$.

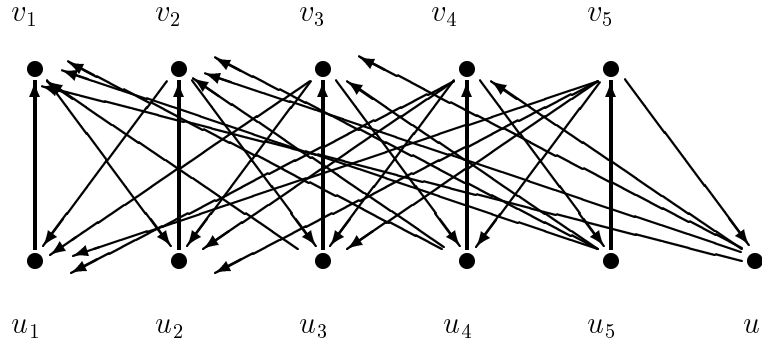


Figure 4: An orientation to reach lower orientable strong radius of $K_{5,6}$.

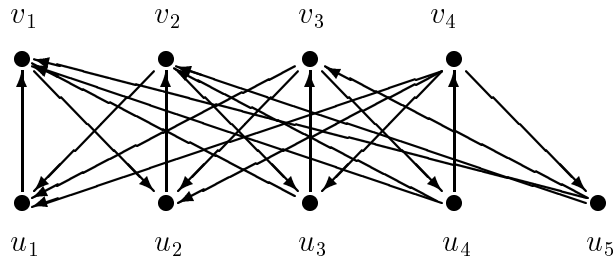


Figure 5: An orientation to reach lower orientable strong radius of and $K_{4,5}$.

Subcase 1: x is in the same partite set of vertices r and s . Let (a_1, x) be an arc on P_2 in D , (b_1, x) be an arc on P_1 on D . Since (s, b_1) must be an arc in D , we have a new path P'_2 , which implies that P_1 and P'_2 have two common edges. Then $\text{sd}(r, s) = |P_1| + |P'_2|$, which is even.

Subcase 2: x is in the other partite set of r and s . Let (y, a_2) be an arc on P_2 , (y, b_2) be an arc on P_1 . Since (b_2, r) must be an arc in D , we have a new path P''_2 , which implies that P_1 and P''_2 have two common edges. Then $\text{sd}(r, s) = |P_1| + |P''_2|$, which is even.

By Case 1 and 2, we know that $\text{sd}(u, v)$ is even for all $u, v \in V(D)$. $\square$

Proposition 1 (From [2]) *For every strong oriented graph D ,*

$$\text{srad}(D) \leq \text{sdiam}(D) \leq 2\text{srad}(D).$$

Theorem 3 *Let $2 \leq m \leq n$. Then*

$$\text{sdiam}(K_{m,n}) = \begin{cases} 4 & \text{if } m = n \\ 6 & \text{if } m \neq n. \end{cases}$$

Proof: Let $V = \{v_1, v_2, \dots, v_m\}$, $U = \{u_1, u_2, \dots, u_n\}$ be two partite sets of $K_{m,n}$.

For $m = n$, since $g(K_{m,n}) = 4$, by Lemma 1, $\text{srad}(D) \geq 4$ for any strong orientation D of $K_{m,n}$. By Proposition 1, such $\text{sdiam}(D) \geq \text{srad}(D)$, $\text{sdiam}(D) \geq 4$. Consider an orientation D_1 such that (u_i, v_i) is an arc in D_1 for $1 \leq i \leq m$, and all other edges are oriente all other edges from V to U . Then $\text{sdiam}(D_1) = 4$ which implies that $\text{sdiam}(K_{m,n}) = 4$.

For $m \neq n$, suppose there exists an orientation D_2 such that $\text{sdiam}(K_{m,n}) = 4$. For any two vertices u_p, u_q in U , $1 \leq p, q \leq n$, there exist $v_r, v_s \in V$, such that $(u_p, v_r), (v_r, u_q), (u_q, v_s)$, and (v_s, u_p) are arcs in D_2 . This implies that it takes at least n vertices in V , which is a contradiction. So $\text{sdiam}(K_{m,n}) > 4$, by Lemma 2, we know that $\text{sdiam}(K_{m,n}) \geq 6$. Consider an orientation D_3 such that (u_i, v_i) is an arc in D_3 for $1 \leq i \leq m$, (u_j, v_m) is an arc in D_3 for $m+1 \leq j \leq n$, and oriente all other edges from V to U . Then $\text{sdiam}(D_3) = 6$, which implies that $\text{sdiam}(K_{m,n}) = 6$. $\square$

Figure 6 and 7 show an orientation to reach lower orientable strong diameter of $K_{4,4}$ and $K_{4,5}$.

Theorem 4 *Let $2 \leq m \leq n$. Then*

$$\text{SDIAM}(K_{m,n}) = \begin{cases} 2m & \text{if } m = n, \\ 2m + 2 & \text{if } m \neq n. \end{cases}$$

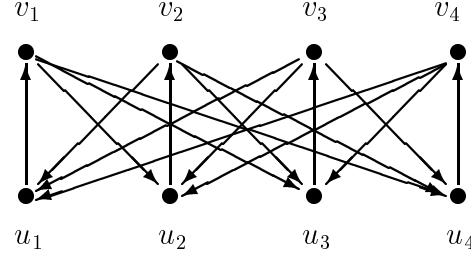


Figure 6: An orientation to reach lower orientable strong diameter of $K_{4,4}$.

Proof: Let $V = \{v_1, v_2, \dots, v_m\}$, $U = \{u_1, u_2, \dots, u_n\}$ be two partite sets of $K_{m,n}$.

For $m = n$, consider an orientation D_1 of $K_{m,n}$. Let (u_m, v_1) be an arc in D_1 , and let $P_1 : v_1, u_1, v_2, u_2, \dots, v_m, u_m$ be a directed path whose length is $2m - 1$. Then $\text{sd}(v_1, u_m) = 2m$. Since every vertex of $K_{m,n}$ has lying on directed path P_1 , we know that P_1 is a longest possible directed path over all orientations of $K_{m,n}$. This implies $\text{sdiam}(D) \leq 2m$ for any orientation D of $K_{m,n}$. Consider an orientation D_2 such that (v_i, u_{i+1}) is an arc in D_2 for $1 \leq i \leq m - 1$, (v_k, u_l) is an arc in D_2 for $2 \leq k \leq m$, $l < k$, and all other edges are assign directed from U to V . Then $\text{sdiam}(D_2) = 2m$ which implies that $\text{SDIAM}(K_{m,n}) = 2m$. An orientation to reach upper orientable strong diameter of $K_{5,5}$ and $K_{4,4}$ are shown in Figure 2 and Figure 3.

For $m \neq n$, Consider an orientation D_3 such that (v_i, u_{i+1}) is an arc in D_3 for $1 \leq i \leq m$, (v_m, u_j) is an arc in D_3 for $m+2 \leq j \leq n$, (v_k, u_l) is an arc in D_3 for $2 \leq k \leq m$, $l < k$, and oriente all other edges from U to V . Then $\text{sdiam}(D_3) = 2m + 2$. Assume that $\text{SDIAM}(K_{m,n}) > 2m + 2$. For any orientation D of $K_{m,n}$, let r, s be two vertices which implies that $\text{sd}(r, s) = \text{sdiam}(D)$.

Case 1: r, s are not in the same partite set. Without loss of generality, let $\text{sd}(v_1, u_n) = \text{sdiam}(D)$. Let (u_n, v_1) be an arc in D , and $P_2 : v_1, u_1, v_2, u_2, \dots, v_m, u_n$ be a directed path whose length is $2m - 1$. Since P_2 is the longest possible directed path in D , we have $\text{sd}(v_1, u_n) = 2m$, which implies that $\text{sdiam}(D) \leq 2m$.

Case 2: r, s are in the same partite set. Without loss of generality, let $\text{sd}(u_1, u_n) = \text{sdiam}(D)$. Let p be the number of vertices in V on $u_1 - u_n$ directed path, q be the number of vertices in V on $u_n - u_1$ directed path. Then $2(p + q) > 2m + 2$, which

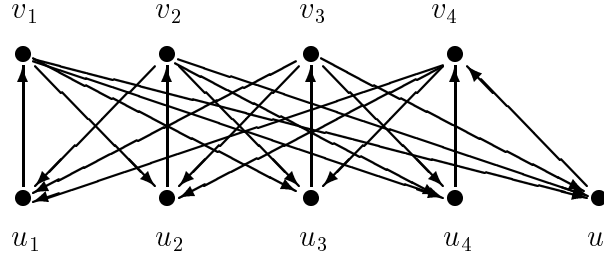


Figure 7: An orientation to reach lower orientable strong diameter of $K_{4,5}$.

implies that $p + 1 > m + 1$. Let P_3 be a directed path from u_1 to u_n , P_4 be a directed path from u_n to u_1 . Let x, y are two vertices in $V(D)$ on P_3 and P_4 .

Subcase 1: Either (u_1, x) or (y, u_n) is an arc in D , and either (u_1, y) or (x, u_n) is an arc in D . Then there exists one vertex (say z) in V on P_3 such that $(u_n, z), (z, u_1)$ are arcs in D . Since the length of the longest possible directed path from u_1 to u_n is $2m$, we have $\text{sd}(u_1, u_n) = 2m + 2$. This implies that $\text{sdiam}(D) \leq 2m + 2$.

Subcase 2: Both (u_1, x) and (y, u_n) are arcs in D , or both (u_1, y) and (x, u_n) are arcs in D . If both (u_1, x) and (y, u_n) are arcs in D , let $P_3 : u_1, x, \dots, y, u_n$ whose length is $2k$. Since $(u_n, x), (y, u_1)$ must be arcs in D , let $P_4 : u_n, x, \dots, y, u_1$ whose length is $2l$. Then $\text{sd}(u_1, u_n) = 2k + 2l - (2l - 2) = 2k + 2$. Since the length of the longest directed path from u_1 to u_n is $2m$, $\text{sd}(u_1, u_n) = 2m + 2$. This implies that $\text{sdiam}(D) \leq 2m + 2$. If both (u_1, y) and (x, u_n) are arcs in D , similarly, we have $\text{sdiam}(D) \leq 2m + 2$.

By Case 1 and Case 2, we know that $\text{SDIAM}(K_{m,n}) = 2m + 2$. $\square$

An orientation to reach upper orientable strong diameter of $K_{5,6}$ and $K_{4,5}$ are shown in Figure 4 and Figure 5.

References

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